

AHA-MEMES: A Fine-Grained Multimodal Benchmark for Understanding Hate in Arabic Memes

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<https://huggingface.co/collections/QCRI/media-integrity-intelligence>

Warning: This paper contains examples that some readers may find potentially sensitive.

Abstract

Hateful memes are a growing form of multimodal online harm, where hostile intent is often conveyed through the joint interpretation of images, text, cultural references, and implicit targets. While hateful meme detection has advanced in high-resource languages, Arabic remains underexplored, with existing meme resources focusing mainly on propaganda or coarse harmful-content labels. We introduce AHA-MEMES,¹ which is, to our knowledge, the first large Arabic hateful meme benchmark with fine-grained, multi-label annotations. The dataset includes 5K manually annotated memes using a taxonomy that captures hate types, i.e., attack strategies. We further provide ~66K silver-labeled memes to support future studies. We benchmark text-only, image-only, and late-fusion multimodal models, as well as few-shot in-context learning (ICL) and open- and closed-weight Vision-Language Models (VLMs) under zero-shot and fine-tuning settings. Our results establish strong baselines and highlight key challenges in culturally grounded Arabic hateful meme detection. We release the dataset, annotation guidelines, and evaluation scripts to support future research.²

1 Introduction

A smiling family photograph may appear harmless on its own, yet the same image can communicate hate when combined with a targeted caption. Likewise, a national flag paired with an apparently benign proverb may become sectarian or xenophobic depending on the communities, histories, and stereotypes that viewers bring to the meme. Memes therefore pose a distinctly multimodal and culturally grounded hate detection problem (Shahroor et al., 2026). Their meaning is often produced through the joint interpretation of image, text, and



Figure 1: Examples from the AHA-MEMES dataset across hateful and non-hateful categories.

shared background knowledge, rather than by either modality in isolation (Sharma et al., 2022). This dependence on cultural context also complicates human judgment, reduces label consistency, and makes automated detection more challenging. In a cross-country study, Bui et al. (2025) found that annotators from five countries agreed on meme labels only 74% of the time, with agreement falling to 67% between U.S. and Indian annotators. Such disagreement among human annotators suggests that models trained mainly on culturally homogeneous and English-centric resources are likely to struggle when hate is expressed through culturally specific visual and linguistic cues. Figure 1 illustrates this with two examples from AHA-MEMES: a doctored portrait that mocks a woman’s appearance (*Hateful*), and a television still repurposed as everyday sarcasm (*Not-Hateful*), where the hateful or benign reading emerges only from the image-text pairing.

Despite growing interest in multimodal hate detection, existing meme benchmarks remain focused in English and often rely on binary or coarse labels (Kiela et al., 2020). This limits their ability to capture *who is targeted*, *what form of hate is*

¹AHA-Memes: Arabic HAteful Memes

²Project resources: [GitHub](#) | [Hugging Face](#)

expressed, and how hateful meaning is constructed through the interaction of image, text, and cultural context. Recent multilingual efforts have begun to broaden coverage, but available resources remain limited in scale and granularity. For example, Multi3Hate provides parallel multilingual annotations but only 300 samples per language (Bui et al., 2025), while GuardHarMem introduces fine-grained meme hate labels only for English (El-amrany et al., 2025). Arabic is especially underserved in this space. Although Arabic text-based hate-speech detection has advanced through shared tasks such as OSACT (Mubarak et al., 2022), and Arabic meme research has studied propaganda in ArMeme (Alam et al., 2024a; Kmainasi et al., 2025a), there is still no dedicated benchmark for fine-grained hate detection in Arabic memes.

We address this gap with AHA-MEMES, a manually annotated fine-grained multimodal dataset for Arabic hateful memes. The dataset includes hateful and non-hateful memes and provides multi-label annotations for hate types. Figure 2 gives an overview of the end-to-end construction pipeline. We conduct detailed experiments on a broad set of systems, including text encoders, image encoders, multimodal fusion models, open-weight VLMs, and closed-weight VLMs and LLMs. Our experiments show that fine-tuned open VLMs perform best for binary detection, that embedded text carries much of the signal while visual information remains complementary, and that fine-grained hate categorisation remains challenging. We also evaluate retrieval-augmented few-shot in-context learning as a training-free alternative. Our contributions are as follows.

- We introduce AHA-MEMES, the first large Arabic hateful-meme dataset, consisting of 5K *manually annotated* memes with binary hate labels and fine-grained multi-label hate types. In addition, we provide silver labels³ for targeted protected categories and features such as dialect, emotion, and sentiment.
- We develop an Arabic-contextual bilingual annotation scheme for binary hate labels, fine-grained hate types, and targeted protected categories, which can support future dataset development for Arabic hateful-meme analysis.
- We provide a comprehensive benchmark across five model families, comparing text-only, image-

only, multimodal fusion, open-weight VLM, and closed-weight VLM/LLM systems under zero-shot, few-shot, and fine-tuning settings.

- We release an additional ~66K Arabic meme corpus with silver labels for binary hate, fine-grained hate type, target category, dialect, emotion, and sentiment, enabling future work on large-scale Arabic multimodal harmful-content modelling.

2 Related Work

Hateful meme detection and resources. The Facebook Hateful Memes Challenge (Kiela et al., 2020) established hateful-meme detection as a multimodal task that requires joint reasoning over image and text. Its use of benign confounders showed that neither modality alone is sufficient, motivating later work on cross-modal fusion, visual-to-text prompting, and LLM-based reasoning for meme interpretation (Kumar and Nandakumar, 2022; Cao et al., 2022; Lin et al., 2024). More recent studies further show that external knowledge, cross-modal alignment, and cultural grounding are important for reliable meme understanding (Sharma et al., 2022; Ren et al., 2026; Kmainasi et al., 2025a). However, most research in this area remains English-centric, and recent evidence suggests that VLMs may inherit culturally narrow assumptions when applied to non-English memes (Wang et al., 2026).

Recent efforts have begun to address under-resourced languages, including Bengali, Tamil, Malayalam (Shahroor et al., 2026). However, these resources remain fragmented in scale, language coverage, and annotation depth (Das and Mukherjee, 2023; Ponnusamy et al., 2024; Lu et al., 2024). In addition, most resources use binary or coarse labels, which obscure *who* is targeted, *what* type of hate is expressed, and *how* hate is rhetorically framed.

Arabic resources. Arabic multimodal disinformation or harmful-content research has been driven mainly by text-based datasets and shared tasks. OSACT advanced offensive-language and hate-speech detection in Arabic social media (Mubarak et al., 2022), while related shared tasks and datasets have addressed propaganda detection and explanation in Arabic and multilingual settings (Hasanain et al., 2025). Recent multilingual and multimodal research on Arabic has addressed fact-checking (Alam et al., 2023) and the understanding of news and social-media content (Kmainasi et al., 2025b). Arabic multimodal social-media analysis has fo-

³By silver labels, we refer to labels generated by an LLM. For this task we used Gemini-3.1-Pro.

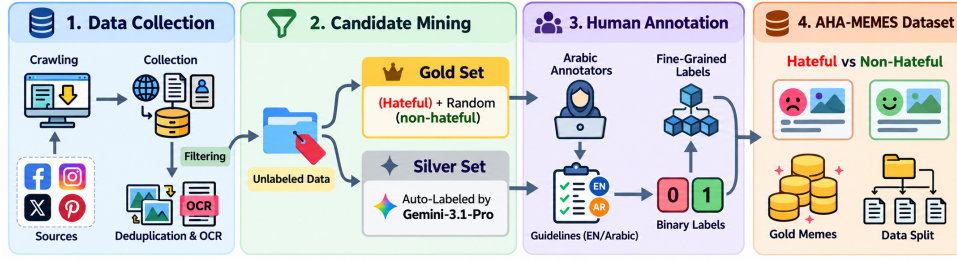


Figure 2: Overview of the AHA-MEMES construction pipeline.

cused primarily on propaganda rather than hate. The ArMeme dataset (Alam et al., 2024b) and AraIEval shared tasks (Hasanain et al., 2024b) introduced Arabic meme propaganda-technique detection, followed by work on coarse hate labels, explanation-enhanced modelling, and explainable Arabic-English propaganda detection (Alam et al., 2024a; Kmainasi et al., 2025a; Alam et al., 2022). These resources are valuable for studying persuasion and propaganda, but they do not provide systematic fine-grained annotations for hateful memes, including hate, and fine-grained hate type, as presented in Table 1. We address this gap by developing AHA-MEMES, a dedicated fine-grained benchmark for Arabic hateful-meme detection.

In-Context learning for multimodality. In-context learning offers a training-free alternative to fine-tuning, however, its effectiveness depends strongly on which demonstrations are included in the prompt (Brown et al., 2020; Abdelali et al., 2024). Similarity-based selection is a common strategy for selecting useful examples in text classification (Liu et al., 2022). In multimodal learning, most ICL work has focused on captioning and visual question answering, often using retrieval from a single modality. It remains underexplored for multimodal *classification*, especially in culturally grounded and low-resource settings. We address this gap by selecting demonstrations through reciprocal-rank fusion of text and image similarities (Cormack et al., 2009).

3 Dataset

3.1 Data Curation

We sourced memes from Facebook, Instagram, Pinterest, and Twitter (X). For Facebook, Instagram, and Pinterest, we manually selected public groups and pages focused on public figures, politics, and social commentary, contexts in which hateful memes are likely to circulate. We then collected images from these public sources using a semi-manual procedure following prior work

(Alam et al., 2024b). For Twitter (X), we collected memes through keyword-based search. In Figure 2, we provide the data curation pipeline. The final 5K set consists of 3,026 Instagram (60.5%), 1,647 Facebook (32.9%), 233 Pinterest (4.7%), and 94 Twitter/X (1.9%) memes. They were posted between April 2016 and March 2025.

3.2 Filtering and OCR Text

Duplicate removal. Social-media collections are highly redundant as users often repost identical or slightly modified content. We therefore removed exact and near-duplicate images after initial data curation. For near-duplicate filtering, we extracted image embeddings using a model fine-tuned on social-media images (Alam et al., 2020) following the approach discussed in (Alam et al., 2024b). We then computed pairwise Euclidean distances between image embeddings across the full dataset. Image pairs with a distance of 3.6 or lower were treated as near-duplicates, and only one image from each duplicate cluster was retained.

OCR text. We extracted embedded text using EasyOCR,⁴ which prior work uses for memes text extraction (Kirk et al., 2021; Cao et al., 2023; Alam et al., 2024b) which we selected for its broad script coverage and scene-text detection capabilities (Liao et al., 2023). As our study focuses on multimodal image-text analysis, we discarded memes with no detectable embedded text. As a result, every retained meme contains both visual and textual modalities.

3.3 Image Selection for Annotation.

Harmful, hateful, and misleading content represents a relatively small proportion of online content, however, its societal impact can be substantial, as reported in prior studies (Sharma et al., 2022). To increase the proportion of relevant examples while preserving coverage of non-hateful memes, we used a weak pre-selection strategy before man-

⁴<https://github.com/JaidedAI/EasyOCR>

Dataset	Lang.	Size	Multi-label	Task granularity	Source
ArMeme (Alam et al., 2024b)	AR	5,725	No	Propaganda status	Social media
ArAIEval (Hasanain et al., 2024b) [†]	AR	3,062	No	Binary propaganda	ArMeme
Propaganda-to-Hate (Alam et al., 2024a) [†]	AR	~3.1K	Yes	Hate + hate types	ArAIEval
MAHED Task 3 (Zaghouani et al., 2025) [†]	AR	3,562	No	Binary hate	Social media
MemeXplain (Kmainasi et al., 2025a) [†]	AR/EN	5,725 AR + 10K EN	No	Label + explanations	ArMeme + Hateful Memes
AraHarMeme (El-Amrany et al., 2026)	AR	5,313	No	Three-way harmfulness	Social media
ArPoMeme (Zaghouani et al., 2026)	AR	~7.3K	Yes	Ideology and polarization	Political memes
AHA-MEMES (ours)	AR	5K + 66K[‡]	Yes	Hate + 8 hate / 3 non-hate types	four platforms

Table 1: Comparison of Arabic multimodal meme datasets. [†]The resource contains data derived from or overlapping with an earlier Arabic meme dataset. [‡]AHA-Memes contains 5K human-annotated gold memes and ~66K auxiliary silver-labeled memes.

ual annotation. Specifically, we applied Gemma-3-12B to assign binary hateful labels to 71K memes.

We then selected 5K memes from this pre-screened pool for manual annotation, including ~2K memes pseudo-labeled as hateful and ~3K sampled uniformly at random from the non-hateful pseudo-labeled subset. After annotation, the resulting dataset contained 1,850 hateful and 3,150 non-hateful memes, providing broad coverage of both classes.

3.4 Annotation

Manual annotation tasks. We manually annotate each meme for binary harmfulness and, fine-grained hierarchical labels. The annotation scheme is organized as follows.

- `is_hateful` is a binary judgment. Following established definitions (Kiela et al., 2020), a meme is labeled as *Hateful* if it directly or indirectly attacks people on the basis of a protected characteristic, including ethnicity, race, nationality, immigration status, religion, caste, sex, gender identity, sexual orientation, disability, or disease. We make two boundary cases explicit. *First*, attacks against groups that themselves promote or perpetrate hate, such as terrorist organizations, are not treated as hate speech. *Second*, content that is offensive, rude, or insulting but does not target a protected category is labeled *Not-Hateful*. This separates harmfulness from generic offensiveness.
- For memes labeled *Hateful*, and `hateful_type` captures the attack strategy as a multi-label annotation over *Mocking*, *Incitement*, *Dehumanization*, *Slurs*, *Contempt*, *Inferiority*, and *Exclusion*. A meme may receive multiple hate-type labels (e.g., *Slurs* and *Dehumanization*).
- For memes labeled *Not-Hateful*, and `non_hateful_type` assigns one or more labels from *Humor*, *Sarcasm*, and *Other*.

This hierarchical structure supports binary hate

detection at the top level, enables fine-grained categorization for hateful content, and ensures that fine-grained labels are consistent with the binary decision by construction. We provide bilingual (Arabic and English) annotation guidelines, including category definitions and examples, in Appendix B.

Annotation team. Annotation was conducted by a third-party company. Three trained native Arabic-speaking annotators annotated the dataset using bilingual guidelines (See Appendix B) in English and Arabic, which we developed to capture dialectal and culturally specific cues (Hasanain et al., 2024a). Before annotation, annotators were informed that the dataset may contain offensive, hateful, or otherwise harmful content. In addition, they were informed about the use of the data. We also conducted multiple rounds of training and guideline refinement to improve annotation consistency and quality. For both the binary and fine-grained tasks, final labels were determined by majority agreement among the annotators. In total, AHA-MEMES contains 5K manually annotated memes, with an annotation cost of ~\$4K.

Annotator agreement. We measured agreement with Cohen’s κ inter-annotator agreement on the annotated dataset. For each multi-label task, every subtype was treated as a separate binary label. Cohen’s κ was computed independently for each subtype and each of the three annotator pairs, then macro-averaged across subtypes and annotator pairs. Agreement is 0.91, 0.75, and 0.67 for `is_hateful`, `hateful_type`, and `non_hateful_type`, respectively, indicating substantial to near-perfect agreement for all annotation tasks (Landis and Koch, 1977).

3.5 Statistics and Data Splits

Distribution. AHA-MEMES is moderately imbalanced, with 1,850 memes (37.0%) labeled as *Hateful*. The fine-grained label distribution is long-tailed (Table 2). On the non-hateful side, *Sarcasm*

(1,393) and *Humor* (1,331) are the most frequent labels. Among hateful memes, *Mocking* (1,007) is the most common attack strategy, whereas *Exclusion* (17) is the rarest. Since fine-grained labels are multi-label, the total label count exceeds the number of memes. The proportion of multi-label memes also increases from 9.9% in the train split to 21.4% in the test split, reflecting the richer co-occurring strategies captured in the triple-annotated gold subset. Per-split label cardinality is reported in Appendix C.3.

Splits. We partition AHA-MEMES into train, development, and test splits containing 3,500 (70%), 500 (10%), and 1,000 (20%) memes, respectively. The splits are stratified by the binary hate label, and we ensure that no meme appears in more than one split. The label distribution across the train, development, and test sets is reported in Table 2.

Hate	Fine-grained	Train	Dev	Test	Total
Hateful	Mocking	706	90	211	1,007
Hateful	Incitement	320	51	85	456
Hateful	Dehumanization	247	42	58	347
Hateful	Slurs	252	42	47	341
Hateful	Contempt	107	18	50	175
Hateful	Inferiority	57	14	32	103
Hateful	Exclusion	10	4	3	17
Hateful	Other (H)	18	2	7	27
Not Hateful	Other (NH)	380	50	96	526
Not Hateful	Sarcasm	934	126	333	1,393
Not Hateful	Humor	863	136	332	1,331

Table 2: Fine-grained sub-type distribution across splits. Counts sum to more than the number of memes because the label is multi-label. *Humor*, *Sarcasm*, and *Other* are non-hateful subtypes; the rest are hateful attack types. H: Hateful; NH: Not-Hateful.

3.6 Auxiliary Silver Dataset

In addition to the 5K human-verified memes that constitute the core AHA-MEMES, we release an auxiliary dataset of $\sim 66K$ Arabic memes with LLM-generated annotations from Gemini 3.1 Pro.⁵ We refer to this resource as a *silver* dataset (Wang et al., 2024; Du et al., 2023) as its labels are not human-verified. The dataset is intended to support weakly supervised learning, semi-supervised learning, active learning, retrieval-based analysis, and future annotation expansion.

The motivation for releasing this auxiliary set is twofold. *First*, even without labels, a large in-

domain collection of Arabic memes is valuable for the community. It captures meme templates, OCR noise, dialectal and code-mixed text, culturally specific references, visual stereotypes, and rapidly evolving political and social topics. Such data can support meme-understanding tasks that require joint reasoning over image content, embedded text, social context, and target entities (Pramanick et al., 2021). *Second*, the silver labels provide a scalable starting point for methods that explicitly model noisy supervision. Weak-supervision frameworks have shown that noisy heuristic labels can be useful when denoised or combined with small amounts of gold data (Ratner et al., 2016, 2017). Similarly, semi-supervised and self-training methods use pseudo-labels on large unlabeled collections to improve model learning, especially when combined with confidence filtering and consistency-based training (Sohn et al., 2020). Recent work on LLM-assisted annotation further suggests that LLMs can reduce annotation cost and help bootstrap labeled resources, while also emphasizing the importance of distinguishing model-generated annotations from human-verified labels (Tan et al., 2024; He et al., 2024).

We therefore position the $\sim 66K$ set as an auxiliary community resource rather than as a core contribution. All primary results in this paper are reported on the human-verified 5K set. Silver annotations are produced with Gemini 3.1 Pro using two prompts: for the unlabelled $\sim 66K$ memes the model predicts the label and subtype together with rich metadata (topic, dialect, sentiment, target categories, propaganda techniques, etc.), whereas for the 5K gold memes it generates the same metadata *conditioned* on the human label, so it never relabels them. The silver labels indicate that hateful content is relatively rare (0.13 hateful and 0.87 non-hateful) while the collection remains dialectally and topically diverse. Full distributions and the prompts are provided in Appendix E.

3.7 Models

We evaluate five model groups spanning unimodal encoders, multimodal fusion models, and vision-language models (VLMs).

Unimodal baselines. Text-only models operate on OCR text and include AraBERTv2 (Antoun et al., 2020), MARBERTv2 (Abdul-Mageed et al., 2021), CAMELBERT-mix (Inoue et al., 2021), XLM-R-base (Conneau et al., 2020), and mBERT (Devlin et al., 2019). Image-only models operate on the

⁵<https://deepmind.google/models/model-cards/gemini-3-1-pro/>

meme image and include ViT-B/16 (Dosovitskiy et al., 2021), BEiT-B/16 (Bao et al., 2022), Swin-B (Liu et al., 2021), ConvNeXt-V2-tiny (Woo et al., 2023), and DINOv2-B (Oquab et al., 2024). Each encoder is followed by a linear classification head.

Late fusion. We combine the two strongest text and vision encoders by concatenating their pooled representations. The resulting vector is passed to a two-layer MLP with GELU activation, a hidden size of 512, and dropout of 0.1.

VLMs. We benchmark open- and closed-weight VLMs, including Qwen3-VL-8B-Instruct (Qwen3-VL-8B) (Bai et al., 2025), Fanar-2-Oryx-IVU (FANAR TEAM et al., 2026), Phi-3.5-Vision (Abdin et al., 2024), SmolVLM-Instruct (Marafioti et al., 2025), Qwen3-VL-8B-Thinking, InternVL3.5-8B (Wang et al., 2025), Gemini-2.5-Pro (Comanici et al., 2025), Gemini-3.5-Flash (Comanici et al., 2025), and GPT-5 (Singh et al., 2026). Each model receives the meme image together with its Arabic OCR text. Unless otherwise stated, all VLM models are evaluated in a zero-shot setup.

Fine-tuning and few-shot evaluation. We use Qwen3-VL-8B to compare supervised fine-tuning with retrieval-based few-shot prompting under a controlled setting, rather than exhaustively evaluating prompting strategies across all VLMs.

3.8 Few-Shot Selection

For each test meme, we retrieve demonstrations from the training set using a CLIP-style dual encoder. Our primary retriever is jina-clip-v2 (Koukounas et al., 2024), which embeds images and text in a shared multilingual space; SigLIP-2 (Zhai et al., 2023) is used as an ablation. Each prompt contains a system instruction, K demonstrations, and the test query. Each demonstration includes the meme image, OCR text, and gold label.

We compare text-only, image-only, and reciprocal rank fusion (RRF) retrieval. Text and image retrieval rank candidates by cosine similarity in their respective embedding spaces. RRF combines the two rankings as $s(d) = \sum_{m \in t, v} \frac{1}{60 + \text{rank}_m(d)}$ (Cormack et al., 2009), avoiding direct comparison of modality-specific similarity scores.

3.9 Training and Model Selection

Encoder and fusion models. All supervised encoder and fusion models are trained with AdamW, weight decay of 0.01, a linear learning-rate schedule, and 6% warmup. Binary hate detection uses

a softmax classification head with cross-entropy loss, while fine-grained hate-type prediction uses sigmoid outputs with binary cross-entropy loss for multi-label classification. For each model family, we tune the learning rate, number of epochs, and batch size. Additional implementation details are provided in Appendix C.2.

VLM fine-tuning. Open-weight Qwen3-VL models are fine-tuned using LoRA with rank 16 and $\alpha=32$, applied to all linear layers while keeping the vision encoder frozen. We train for 3 epochs with a learning rate of 1×10^{-4} , a cosine schedule, and bfloat16 precision.

Model selection and evaluation. For supervised experiments, models are trained on train, selected on dev by macro-F1, and evaluated once on the held-out test set using the best configuration. For fine-grained hate-type prediction, we additionally tune the sigmoid decision threshold on dev. Zero-shot VLMs are evaluated directly on test.

4 Results and Discussion

4.1 Results Across Model Families

In Table 3, we report performance across all experimental settings including random and majority-class baselines.

Fine-tuned VLMs are strongest for binary task.

Qwen3-VL-8B with LoRA achieves the best binary macro-F1 of 0.768. It outperforms the best late-fusion model, the best zero-shot closed VLM, and the strongest text encoder. Qwen3-VL-2B also performs well, reaching 0.739 macro-F1, which is close to the best fusion models. This suggests that supervised multimodal models are the most effective approach when labeled data are available.

Zero-shot VLMs show low hateful-class recall.

Qwen3-VL-8B-Instruct is the strongest zero-shot open VLM, reaching 0.643 binary macro-F1, followed by Fanar-2-Oryx-IVU. The remaining open VLMs perform close to the majority baseline on binary and show very low *Hateful* recall. The best zero-shot open VLM falls behind Gemini-2.5-pro by about 7 macro-F1 and the fine-tuned Qwen3-VL-8B by about 13 points. Fine-tuning substantially changes this pattern, improving Qwen3-VL-8B from 0.643 to 0.768 on binary and from 0.176 to 0.334 on fine-grained hate-type prediction.

Fine-grained hate-type classification remains difficult. The best fine-grained hate-type macro-F1 is 0.340, achieved by Gemini-2.5-pro, with fine-tuned Qwen3-VL-8B close behind at 0.334. All

Model	Binary			Fine-grained	
	Ma-F1	Acc.	Rec. _H	Ma-F1	Mi-F1
Baselines					
Majority	0.399	0.663	0.000	0.050	0.295
Random	0.500	0.553	0.337	0.125	0.233
Text (FT)					
MARBERTv2	0.709	0.743	0.596	0.263	0.390
CAMELBERT-mix	0.689	0.726	0.567	0.252	0.375
AraBERTv2	0.678	0.706	0.611	0.295	0.411
XLM-R-base	0.650	0.716	0.418	0.259	0.415
mBERT	0.637	0.667	0.564	0.245	0.364
Image (FT)					
ConvNeXt-V2-tiny	0.657	0.700	0.513	0.246	0.418
Swin-base	0.657	0.686	0.585	0.258	0.365
BEiT-base	0.651	0.708	0.451	0.255	0.385
ViT-base	0.645	0.708	0.427	0.268	0.424
DINOv2-base	0.589	0.681	0.309	0.233	0.349
Fusion (FT)					
MARBERTv2+ViT	0.730	0.757	0.650	0.274	0.428
MARBERTv2+BEiT	0.724	0.750	0.656	0.318	0.444
AraBERTv2+ViT	0.706	0.741	0.591	0.279	0.416
AraBERTv2+BEiT	0.698	0.739	0.549	0.298	0.426
Open VLMs (zero-shot)					
Qwen3-VL-8B-Instruct	0.643	0.743	0.318	0.176	0.352
Fanar-2-Oryx-IVU	0.535	0.700	0.154	0.115	0.288
Phi-3.5-Vision	0.486	0.682	0.095	0.044	0.159
SmolVLM-Instruct	0.473	0.559	0.229	0.152	0.341
Qwen3-VL-8B-Thinking	0.469	0.676	0.077	0.191	0.382
InternVL3.5-8B	0.458	0.676	0.062	0.164	0.369
Open VLMs (FT)					
Qwen3-VL-8B	0.768	0.797	0.656	0.334	0.475
Qwen3-VL-2B	0.739	0.782	0.558	0.284	0.437
Open VLM (few-shot)					
Qwen3-VL-8B	0.708	0.771	0.454	0.241	0.405
Closed VLMs (zero-shot)					
Gemini-2.5-Pro	0.711	0.774	0.457	0.340	0.465
GPT-5	0.628	0.741	0.282	0.301	0.464
Gemini-3.5-Flash	0.499	0.694	0.104	0.271	0.486

Table 3: Results on the AHA-MEMES test split. FT denotes fine-tuning. Ma-F1 and Mi-F1 denote macro-F1 and micro-F1, respectively. Rec._H denotes recall for the *Hateful* class. Fine-grained macro-F1 is averaged over all ten categories. Bold indicates the best result per column.

model groups remain below 0.35 macro-F1. The gap between macro-F1 and micro-F1 suggests that models perform better on frequent labels such as *Humor* and *Sarcasm* than on rare hate types. Fine-tuning improves the performance, however, it does not fully mitigate the effect of label imbalance.

Fusion helps, but text provides the strongest uni-modal signal. Late fusion improves over the best text-only encoder on binary and increases *Hateful* recall, with MARBERTv2+BEiT matching the highest recall of 0.656. Image-only encoders are competitive, however, they are less reliable as standalone models. This pattern suggests that embedded Arabic text carries much of the task signal,

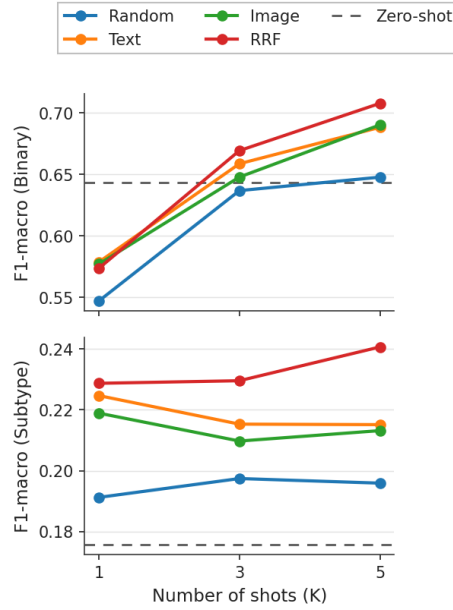


Figure 3: Few-shot learning for Qwen3-VL-8B. Binary performance improves with more samples, however, it falls below zero-shot at $K=1$. Fine-grained hate-type performance improves modestly and saturates early. RRF is the strongest retrieval strategy.

while visual content provides complementary context that is most useful when combined with text.

Arabic-specific encoders provide strong text baselines. Among text-only models, MARBERTv2 achieves the best binary macro-F1, while AraBERTv2 achieves the best fine-grained hate-type macro-F1. Both Arabic-specific encoders outperform mBERT and XLM-R-base on binary, highlighting the value of Arabic-centric pre-training for dialect-rich meme text.

4.2 Effect of Prompting

We compare *direct prompting* with *rationale-style prompting* in English and Arabic for closed VLMs. Full results are reported in Appendix Table 8. Two trends are consistent. *First*, model choice has a larger effect than prompt format. Gemini-2.5-pro performs best across all prompting settings, followed by GPT-5 and Gemini-3.5-flash. *Second*, rationale-style prompting yields limited and model-dependent gains. It has little effect on binary performance, improves fine-grained hate-type prediction for the larger models, and hurts Gemini-3.5-flash when rationales are requested in Arabic. This suggests that *Arabic rationale generation can introduce additional errors* for lower-capacity models.

4.3 Few-Shot ICL

We evaluate retrieval-augmented few-shot ICL. For each test meme, we retrieve K labeled examples from the train split and place them before the query as image-text demonstrations. We compare random demonstrations with text-based retrieval, image-based retrieval, and reciprocal rank fusion (RRF) over jina-clip-v2 embeddings (Koukounas et al., 2024; Cormack et al., 2009). We evaluate $K \in 1, 3, 5$ using Qwen3-VL-8B and Qwen3-VL-2B. Details results are reported in Appendix D.

Few-shot ICL gives moderate gains over zero-shot prompting, but remains below fine-tuning.

As shown in Figure 3, RRF is the strongest retrieval strategy. The best setting is Qwen3-VL-8B with RRF and $K=5$, which improves binary macro-F1 from 0.643 to 0.708 and fine-grained hate-type macro-F1 from 0.176 to 0.241 over the zero-shot Qwen3-VL-8B baseline. However, this remains below full fine-tuning, which reaches 0.768 binary macro-F1 and 0.334 fine-grained hate-type macro-F1. These results show that retrieval selects more informative demonstrations but still falls short of supervised adaptation.

Few-shot mainly improves hateful-class recall.

RRF with $K=5$ raises *Hateful* recall from 0.318 to 0.454, reducing the zero-shot model’s bias toward non-hateful predictions. Gains for fine-grained hate-type prediction are smaller and are driven mostly by frequent labels such as *Humor* and *Mocking*, while rare hateful types remain difficult. A single demonstration is unstable and hurts binary performance, whereas $K=3$ and $K=5$ provide more reliable context.

4.4 Error Analysis

Low recall is the main zero-shot failure. Zero-shot VLMs show low *Hateful* recall on binary. They under-predict the *Hateful* class and therefore miss many hateful memes, making this the most important safety failure for moderation. Fine-tuning mitigates this problem, Qwen3-VL-8B and the best fusion model recover roughly two-thirds of hateful memes. This suggests that the low-recall pattern is mainly a limitation of the zero-shot setting rather than a limitation of the model family.

Fine-grained hate-type predictions are biased toward frequent labels. The zero-shot closed VLMs predict fine-grained hate-type distributions that differ considerably from the gold label distribution, as shown in Appendix Table 9. Gemini-3.5-flash over-

predicts frequent labels such as *Humor* and *Sarcasm*, which improves micro-F1, however, weakens macro-F1. Gemini-2.5-pro over-predicts hateful types such as *Mocking*, *Contempt*, and *Slurs*, increasing recall at the expense of precision. GPT-5 is more cautious. *Exclusion* has only three test instances and is almost never predicted correctly.

Longer meme text increases difficulty. Binary errors increase with the length of the extracted Arabic text for all models, as shown in Appendix Table 10. The trend is steepest for Gemini-3.5-flash. Longer text often require more context, pragmatic inference, and target identification, whereas memes with short text are more likely to contain direct lexical cues. Larger models are more robust, however, the trend remains visible across systems.

Implicit hate is harder to classify. Qualitative inspection of Gemini-2.5-pro with Arabic rationale prompting shows that false negatives are not caused by OCR or image-understanding failures. The model identifies the visual template and reads dialectal Arabic correctly but treats implicit sectarian, religious, or misogynistic attacks as humor or satire and predicts *Not-Hateful*. False positives are offensive dialectal jokes that annotators judged not to be hateful. These cases show that the hardest boundary is the normative judgment of implicit hate rather than multimodal prediction. This motivates the explicit taxonomy and annotation guidelines described in Section 3.4.

5 Conclusion and Future Work

We presented AHA-MEMES, a fine-grained Arabic hateful-meme benchmark comprising 5K human-annotated memes with a hierarchical annotation scheme covering hatefulness and attack subtype. We also release a separate auxiliary corpus of ~ 66 K silver-labeled memes with model-generated labels and metadata for weak supervision rather than evaluation. Experiments show that fine-tuning an open-weight VLM gives the best binary performance, while late fusion and Arabic text encoders remain strong baselines. Text provides the strongest unimodal signal, while visual context offers complementary information. Fine-grained classification remains challenging because models struggle to recover rare and overlapping hate categories. Zero-shot VLMs make conservative predictions and often miss hateful memes, while few-shot prompting provides only modest gains. Future work will focus on rare and co-occurring

hate categories, target-aware modeling, clearer separation between offensive humor and protected-group hate, and broader coverage of dialects and emerging meme formats.

Limitations

AHA-MEMES is designed as a focused benchmark for Arabic hateful-meme understanding rather than a comprehensive representation of Arabic meme culture. We collected data from four public platforms, Facebook, Instagram, Pinterest, and Twitter/X, which gives broad public coverage, however, our collection may not capture all dialects, regions, platform communities, or private and ephemeral content. Some annotation boundaries are inherently nuanced, especially between offensive humor, satire, political criticism, and protected-group hate. We provide detailed annotation guidelines and instruct annotators to be mindful of such issues. The auxiliary silver dataset broadens the resource and supports weakly supervised research; however, it should be used with caution.

Ethics and Broader Impact

AHA-MEMES supports research on Arabic multimodal hate detection and safer content moderation. It is developed from publicly available memes and includes hateful, offensive, and sensitive material solely for research and evaluation. We do not collect any personal information about users or annotators, hence, there is little to no privacy risk. The memes come from public pages and groups and largely depict public figures; we make no attempt to identify depicted individuals. As models can miss harmful content or over-flag legitimate content, they should be used with human oversight, clear documentation, and ongoing monitoring. Annotation was performed by trained, fairly compensated native-speaker annotators, informed in advance about the nature of the content, using guidelines that distinguish protected-group hate from generic offensiveness and legitimate criticism.

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A Data and Resource Release

We made the scripts and dataset available under the CC BY-NC-SA 4.0 license for reproducibility and future extensions.⁶

B Annotation Guidelines

B.1 English Annotation Guideline

Purpose. The purpose of this annotation is: (i) to identify whether a meme is *Hateful* or *Not-Hateful*; (ii) for hateful memes, to identify the attack type(s) (multi-label) and the targeted protected category(ies) (multi-label); and (iii) for non-hateful memes, to assign one or more non-hateful subtypes (multi-label). The annotation design is *hierarchical* and *conditional*: the fine-grained tasks are completed based on the binary label decision, discussed below.

Definitions. We define *hateful* content as a direct or indirect attack on people on the basis of a protected characteristic, including ethnicity, race, nationality, immigration status, religion, caste, sex, gender identity, sexual orientation, and disability or disease. An *attack* includes violent or dehumanizing speech, such as comparing people to non-human entities, statements of inferiority, calls for exclusion or segregation, or mockery of a hate crime. We apply two boundary rules:

- Attacking groups that themselves promote or carry out hate, such as terrorist organisations, is *not* considered hate speech.
- A meme that is insulting or offensive but does not target a protected category is labelled *Not-Hateful* under this scheme.

B.1.1 Task 1: Binary Classification (is_hateful; Hateful vs. Not-Hateful)

The task is to select exactly one of *Hateful* or *Not-Hateful*. The choice determines the subsequent annotation task: if *Hateful*, complete Tasks 2 if *Not-Hateful*, complete Task 3.

B.1.2 Task 2: Attack Type (hateful_type; Hateful only; multi-label)

The task is to select *all* attack types that apply; a meme may exhibit several at once (e.g., *Slurs* + *Dehumanization*, or *Mocking* + *Inferiority*).

- **Dehumanization:** explicitly or implicitly describing a group as subhuman, such as by comparing an ethnic group to animals or using “fish season” metaphors (Figure 4).
- **Inferiority:** claiming that a group is inferior, less worthy, or less important than society or another group, such as through class-based belittling of “peasants” (Figure 5).
- **Incitement:** explicitly or implicitly calling for harm to be inflicted on a group, including physical attacks, such as through imagery of forced physical discipline (Figure 6).
- **Mocking:** joking about, undermining, belittling, or disparaging a group, such as by portraying its members as buffoonish or body-shaming a gender (Figure 7).
- **Contempt:** expressing intensely negative feelings toward a group, such as through statements like “Men are trash” or expressions of ideological contempt (Figure 8).
- **Slurs:** using prejudicial terms to refer to or characterize a group, including racial, religious, or gender-based slurs in Arabic dialects (Figure 9).
- **Exclusion:** advocating, planning, or justifying the exclusion or segregation of a group, such as through statements like “Go back to your country” (Figure 10).
- **Other:** a hateful attack that does not clearly fall under any of the types above. This option is selected only when no other type applies and cannot be combined with another attack type.

B.1.3 Task 3: Not-Hateful Subtype (non_hateful_type; Not-Hateful only; multi-label)

Non-hateful memes contain humorous, neutral, or positive content that does not target or harm a protected group, are intended for entertainment, and do not promote violence, hatred, or discrimination. The task is to select one or more subtypes.

- **Humor:** content intended to entertain or amuse through jokes, puns, exaggeration, or playful commentary on everyday situations (work, family, food, travel), with no negative implication toward a protected group (Figure 11).
- **Sarcasm:** irony in which the intended meaning is the opposite of what is stated, used for humorous effect and directed at general behaviour or institutions (e.g., government, bureaucracy, inflation) rather than a protected group (Figure 12).
- **Other:** neutral, positive, or informational memes that are not clearly humour or sarcasm (e.g., mo-

⁶<https://github.com/MohamedBayan/AHA-Memes>

tivational quotes, announcements, generic reactions) and do not target a protected group.

B.2 Arabic Annotation Guideline

The following is the Arabic version of the guidelines used during annotation.

Purpose and definitions.

الهدف من التصنيف: يقوم المصنّف بالمهام التالية: أولاً، تحديد ما إذا كان الميم يحتوي على خطاب كراهية (Hateful) أم لا (Not-

Hateful). إذا صُنِفَ الميم على أنه يحتوي على كراهية، يختار المصنّف نوع الهجوم ويحدّد الفئة أو الفئات المستهدّفة؛ وقد يتضمّن الميم أكثر من نوع هجوم أو يستهدف أكثر من فئة في آنٍ واحد. أمّا إذا صُنِفَ الميم على أنه لا يحتوي على كراهية، فيختار المصنّف التصنيف الفرعي المناسب.

التعريفات: خطاب الكراهية (Hateful): هو أيّ محتوى يتضمّن هجومًا مباشرًا أو غير مباشر على أشخاص أو مجموعات بسبب خصائص محميّة، تشمل: العرق، الإثنية، الجنسية، وضع الهجرة، الدين، الطائفة، الطبقة الاجتماعية، الهوية الجنسية، التوجه الجنسي، أو الإعاقة أو المرض. يشمل الهجوم: استخدام لغة عنيفة أو مسيئة، أو تجريد الأفراد والجماعات من إنسانيتهم بتشبيههم بالحيوانات أو الجمادات، أو توظيف عبارات تحقير أو دونية، أو الدعوة إلى العزل أو الإقصاء، فضلًا عن الاستهزاء بجرائم الكراهية.

ملاحظات مهمّة: الهجوم على جماعات تمارس الكراهية أو الإرهاب (كالتنظيمات الإرهابية) لا يُعدّ خطاب كراهية وفق هذا التعريف. كذلك، إذا كان الميم مسيئًا أو مُهينًا لكنه لا يستهدف فئة محميّة بعينها، فيُصنّف على أنه غير كراهية (Not-Hateful).

Task 1: Meme classification.

المهمّة ١: تصنيف الميم. حدّد ما إذا كان الميم يحتوي على خطاب كراهية (Hateful) أم لا (Not-Hateful). إذا كان يحتوي على كراهية، انتقل إلى المهمّة ١.٢ والمهمّة ٢.٢. وإذا كان لا يحتوي على كراهية، انتقل إلى المهمّة ٣.

Task 2: Attack types (multi-label).

المهمّة ١.٢: نوع الهجوم (متعدّد التصنيفات). اختر جميع أنواع الهجوم المنطبقة على الميم؛ فقد يجمع الميم الواحد بين أكثر من نوع (مثلًا: ألفاظ مهينة + تجريد من الإنسانية).

١ تجريد من الإنسانية (Dehumanization): وصف مجموعة بشرية صراحةً أو ضمناً بأنها دون البشر، كتشبيه فئة عرقية بالحيوانات أو استخدام تعبيرات مثل «موسم الصيد».

٢ الدونية (Inferiority): الزعم بأن مجموعة ما أقلّ شأنًا أو قيمةً من غيرها، كالتحقير على أساس الطبقة الاجتماعية أو المنطقة الجغرافية.

٣ التحريض على العنف (Incitement): الدعوة صراحةً أو ضمناً إلى إيذاء مجموعة، بما يشمل التحريض على العنف الجسدي.

٤ الاستهزاء (Mocking): السخرية من مجموعة أو التقليل من شأنها أو الحظ من قدرها، كتصوير فئة بصورة مضحكة أو التمرّ على أساس الجنس أو المظهر.

٥ الازدراء (Contempt): التعبير عن مشاعر كراهية أو احتقار شديد تجاه مجموعة بأكملها.

٦ الألفاظ النابية والشتائم (Slurs): توظيف ألفاظ تمييزية أو شتائم مرتبطة بهويّة المجموعة المستهدّفة (عرقية أو دينية أو جنسية).

٧ الإقصاء (Exclusion): الدعوة إلى طرد مجموعة أو عزلها أو فصلها عن المجتمع، كعبارة «ارجعوا لبلادكم».

٨ أخرى (Other): هجوم كراهية لا ينطبق عليه أيّ من الأنواع السابقة بوضوح. يُستخدم فقط حين لا يمكن اختيار نوع آخر، ولا يُجمع مع أنواع أخرى.

C Experimental Details

C.1 Zero-Shot Prompting Setup

Each zero-shot query combines a *system* message and an *instruction* message. The system message frames the model as an expert Arabic multimodal content-moderation annotator covering MSA and the Egyptian, Levantine, Gulf, and Maghrebi dialects. The instruction provides the meme image and OCR text and restricts the output to the pre-defined label space. For binary classification, the model predicts *Hateful* or *Not-Hateful* from the combined image-text content, guided by definitions distinguishing identity-based hate from general offensiveness. It returns the prediction as "label": For fine-grained classification, the prompt lists the ten taxonomy labels with brief definitions and asks for applicable labels as a JSON list. The COT-EN and COT-AR variants add a free-text reasoning field in English or Arabic, respectively. We fine-tune the open-weight VLMs with the same binary and fine-grained templates to generate the target label set directly.



Figure 4: Example memes annotated as *Hateful / Dehumanization*.



Figure 5: Example memes annotated as *Hateful / Inferiority*.



Figure 6: Example memes annotated as *Hateful / Incitement*.

C.2 Training and Compute

We fine-tune the text, image, and fusion models with the HuggingFace Transformers Trainer.

We use AdamW with weight decay 0.01, a linear learning-rate schedule, 6% warmup, and 128-token text inputs. Hyperparameters are selected from the



Figure 7: Example memes annotated as *Hateful / Mocking*.



Figure 8: Example memes annotated as *Hateful / Contempt*.

search space as reported in Table 4, and the best checkpoint is chosen based on macro-F1 on the development set. For fine-grained hate-type prediction, we tune a sigmoid threshold on the development set over 0.05, 0.10, ..., 0.90. Open-weight Qwen3-VL models are fine-tuned with ms-swift using LoRA with rank 16 and $\alpha=32$, across all lin-

ear layers with the vision encoder frozen. We train for 3 epochs with learning rate 1×10^{-4} , a cosine schedule, and bfloat16. All experiments use seed 42 on NVIDIA H200 GPUs.



Figure 9: Example memes annotated as *Hateful / Slurs*.



Figure 10: Example annotated as *Hateful / Exclusion*.

Hyper-param.	Text	Image	Fusion	VLM
Learning rate	2,3,5 $\times 10^{-5}$	3,5 $\times 10^{-5}$	2,3 $\times 10^{-5}$	1 $\times 10^{-4}$
Epochs	5, 10	10, 15	5, 10	3
Batch size	16, 32	32	16	4

Table 4: Grid-search space for the fine-tuned families (best by dev macro-F1). The open VLMs use fixed LoRA settings rather than a grid. The fusion-head learning rate is fixed at 1×10^{-4} . All runs use seed 42.

C.3 Additional Dataset Analysis

OCR-text length. In Table 5, we report the distribution of OCR-extracted text length. The splits are well matched (mean ≈ 86 –88 characters), and *Hateful* memes carry noticeably more text than *Not-Hateful* ones on the test split (96.2 vs. 83.1 characters), consistent with the finding (Section 4.4) that hateful intent is often conveyed through longer, more contextual captions.

Multi-label distribution. In Table 6, we report the

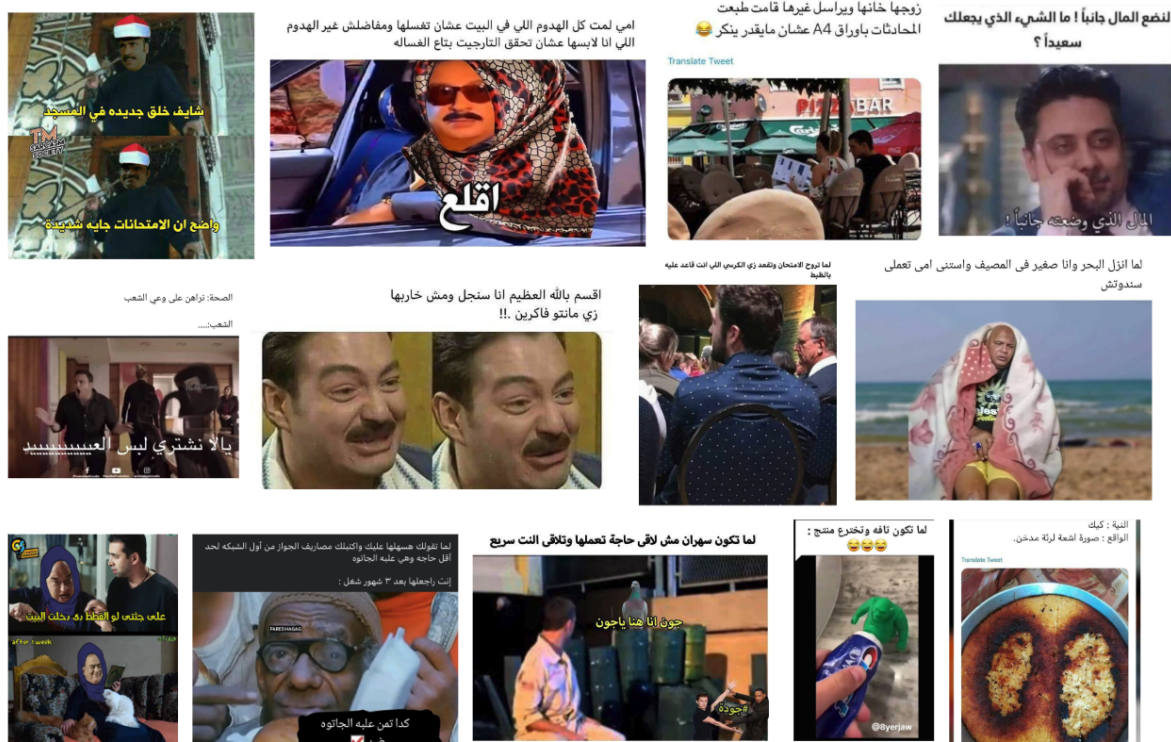


Figure 11: Example memes annotated as *Not-Hateful / Humor*.



Figure 12: Example memes annotated as *Not-Hateful / Sarcasm*.

number of fine-grained categories assigned to each meme. The test split shows the greatest category overlap, with an average of 1.25 labels per meme; 21.4% of test memes receive at least two categories, compared with 9.9% in the training split.

Label co-occurrence. Table 7 shows the most fre-

quent fine-grained label pairs. *Mocking* frequently co-occurs with *Dehumanization*, *Incitement*, *Contempt*, and *Slurs*, while *Humor* and *Sarcasm* often co-occur among non-hateful memes. These patterns support modeling hate as combinations of strategies rather than assigning a single fine-

grained label per meme.

Split	Mean	Median	Max
Train	85.5	76	641
Dev	87.8	80	619
Test	87.5	77	644

Table 5: OCR-text length (characters) per split. *Test set class-wise*: Hateful 96.2; Not-Hateful 83.1 (mean)

Split	Avg	1	2	3	4
Train	1.11	3,154	298	48	0
Dev	1.15	439	47	14	0
Test	1.25	786	178	32	4

Table 6: Fine-grained label cardinality: number of memes with 1-4 fine-grained labels, and the mean number of labels per meme.

Fine-grained pair	Count
Dehumanization + Mocking	145
Incitement + Mocking	114
Humor + Sarcasm	81
Contempt + Mocking	78
Dehumanization + Incitement	66
Mocking + Slurs	60
Inferiority + Mocking	55
Incitement + Slurs	44
Contempt + Incitement	39
Dehumanization + Slurs	31

Table 7: Most frequent fine-grained hate-type co-occurrence pairs (all splits).

C.4 Closed-Weight VLM Prompting

Table 8 presents the full prompting comparison summarized in Section C.5. We compare closed-weight VLMs using direct prompting and chain-of-thought reasoning in English and Arabic. The relative model ranking remains largely consistent across prompting strategies. Chain-of-thought improves fine-grained performance for the stronger models, whereas Arabic reasoning reduces performance for Gemini-3.5-Flash.

C.5 Additional Error Analysis

We provide additional analyses to support the zero-shot error patterns discussed in Section 4.4. Table 9 compares the number of predictions for each fine-grained label with its gold frequency across closed-weight VLMs. The results show systematic over-prediction of common non-hateful labels, particularly *Humor* and *Sarcasm*, as well as several hateful categories. Table 10 further examines binary errors by OCR-text length. Error rates increase for longer captions across all models, with

Model	Prompt	Binary	Fine-grained
Gemini-2.5-pro	DIRECT	0.711	0.340
	COT-EN	0.709	0.358
	COT-AR	0.726	0.333
GPT-5	DIRECT	0.628	0.301
	COT-EN	0.642	0.313
	COT-AR	0.647	0.315
Gemini-3.5-flash	DIRECT	0.499	0.271
	COT-EN	0.499	0.272
	COT-AR	0.493	0.265

Table 8: Macro-F1 of the zero-shot closed VLMs under direct prompting (DIRECT; prediction without explicit reasoning) and chain-of-thought prompting in English (COT-EN) or Arabic (COT-AR). The best value per column is in bold.

the largest degradation observed for the smallest model.

Label	Gold	Pro	GPT-5	Flash
Humor	332	728	878	885
Sarcasm	333	349	404	869
Mocking	211	465	164	120
Contempt	50	264	64	45
Slurs	47	166	31	9

Table 9: Subtype label over- and under-prediction (predicted count vs. gold support) for the zero-shot closed models under the DIRECT prompt.

D Few-Shot Results

In Tables 11 and 12, we report the strategy-by- K results for Qwen3-VL-8B and Qwen3-VL-2B, respectively. For Qwen3-VL-8B, RRF achieves the best performance on both subtasks and improves consistently up to $K=5$. On binary task, however, all strategies perform below zero-shot at $K=1$. Qwen3-VL-2B follows a different pattern, peaking at $K=3$ and dropping at $K=5$, while image-only retrieval matches RRF. These results suggest that the smaller model benefits from a small number of demonstrations, although it uses longer or fused demonstration contexts less effectively.

Model scale. In Figure 13, we compare the best F1 over $K \in \{1, 3, 5\}$ for each retrieval strategy and model size. Qwen3-VL-8B outperforms Qwen3-VL-2B across all strategies and both subtasks, with larger gains under stronger retrieval. Moving from random retrieval to RRF improves Qwen3-VL-8B from 0.650 to 0.708 on binary and from 0.203 to 0.241 on fine-grained hate-type prediction, while Qwen3-VL-2B gains less. RRF performs best for both models, however, only the 8B

Model (DIRECT)	21–60	61–120	120+
Gemini-2.5-pro	0.222	0.221	0.255
GPT-5	0.239	0.260	0.299
Gemini-3.5-flash	0.258	0.312	0.382

Table 10: Binary error rate by OCR-text length (characters). Longer, more contextual captions are harder, and the smallest model degrades most.

Strategy	Binary (macro-F1)				Fine-grained (macro-F1)		
	$K0$	$K1$	$K3$	$K5$	$K1$	$K3$	$K5$
zero-shot	0.643	—	—	—	—	—	—
random	—	0.547	0.637	0.648	0.191	0.197	0.196
text	—	0.579	0.659	0.688	0.225	0.215	0.215
image	—	0.577	0.648	0.690	0.219	0.210	0.213
RRF	—	0.574	0.669	0.708	0.229	0.230	0.241

Table 11: Retrieval-augmented few-shot ICL on **Qwen3-VL-8B** (Fine-grained zero-shot = 0.176). RRF beats random and, at $K \geq 3$, zero-shot. Best per task in bold.

Strategy	Binary			Fine-grained		
	$K1$	$K3$	$K5$	$K1$	$K3$	$K5$
random	0.590	0.623	0.615	0.131	0.140	0.145
text	0.639	0.658	0.629	0.175	0.161	0.189
image	0.661	0.681	0.665	0.160	0.210	0.174
RRF	0.663	0.684	0.671	0.203	0.181	0.209

Table 12: Retrieval-augmented few-shot ICL on **Qwen3-VL-2B**. The small model follows an inverted-U pattern, rising up to $K=3$ and then falling at $K=5$, unlike the 8B.

model consistently benefits from the fused retrieval signal. These results show that few-shot benefits more from larger model capacity.

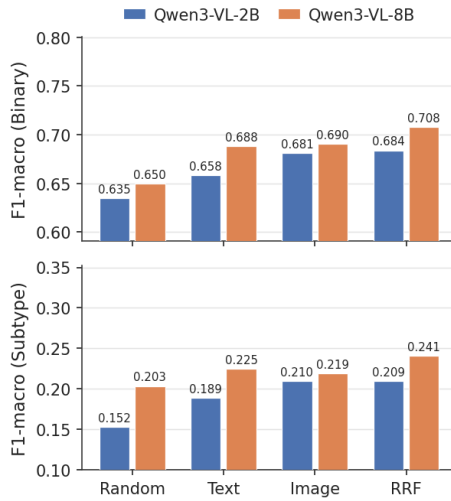


Figure 13: Best F1 (max over $K \in \{1, 3, 5\}$) per retrieval strategy for Qwen3-VL-2B vs. 8B. The 8B wins at every strategy; the gain from random to RRF is larger for the 8B, on both Binary (top) and Subtype (bottom).

E Details of Silver Dataset

Selection and silver labels. After removing duplicates and filtering out memes without extracted text, we retain $\sim 71K$ memes. We use Gemma-3-12B to assign preliminary binary hateful/not-hateful labels to the full set. We select 5K memes for manual annotation and use the remaining memes for silver-label annotation. Prompts are listed in Section E.1.

Silver-label distributions. Tables 13 and 14 summarize the Gemini-3.1-Pro silver labels for the 66,297 auxiliary memes. The set is dominated by benign, humorous content, with less than 1% labeled as hateful. Despite this skew, the memes remain multimodally challenging, as about one quarter convey implicit meaning and most rely on complementary text-image information. The collection is also predominantly in Egyptian Arabic and frequently reflects local, regional, or national context. Everyday life is the dominant topic, followed by entertainment and culture, while *Gender/Sex* is the most frequent mentioned target. Overall, the silver set is largely humorous and non-hateful, yet strongly contextual and multimodal.

E.1 Prompts

We provide the prompts used for binary hate detection, fine-grained hate-type prediction, and silver labelling of the unlabelled set. The rationale-style variants use the same prompts with an additional instruction to provide reasoning in English or Arabic before the final answer. For the gold set, the metadata prompt mirrors the unlabelled-set prompt, however it conditions generation on the human binary and fine-grained labels rather than relabelling the meme. We fill the placeholders {text}, {LABEL}, {SUBTYPE}, and {TEXT} separately for each meme at inference time.

Task 1: Binary hatefulness classification.

```
# ===== SYSTEM PROMPT =====
You are an expert content-moderation annotator
specializing in Arabic multimodal memes,
covering Modern Standard Arabic and
regional dialects (e.g., Egyptian,
Levantine, Gulf, Maghrebi).

Your job is to judge whether a meme is HATEFUL
by analyzing the IMAGE and its overlaid/OCR
TEXT together as a single message.
Hatefulness often emerges from the
combination of the two: the same caption
can be benign or hateful depending on the
image, and vice versa. Account for sarcasm,
irony, visual symbols, well-known figures,
```

HATEFULNESS					
Not-Hateful	65,717 (99.1%)	Hateful	580 (0.9%)		
FINE-GRAINED CATEGORY					
Humor	60,406 (91.1%)	Sarcasm	2,884 (4.4%)	Other	2,429 (3.7%)
Mocking	335 (0.5%)	Slurs	91 (0.1%)	Contempt	55 (0.1%)
Dehumanization	47 (0.1%)	Incitement	33 (<0.1%)	Inferiority	9 (<0.1%)
Exclusion	8 (<0.1%)				
PROPAGANDA					
Absent	64,375 (97.1%)	Present	1,922 (2.9%)		
SENTIMENT					
Neutral	32,329 (48.8%)	Positive	26,147 (39.4%)	Negative	7,821 (11.8%)
STANCE					
Neutral	53,956 (81.4%)	Mocking	8,421 (12.7%)	Supportive	2,210 (3.3%)
Critical	1,686 (2.5%)	Ambiguous	24 (<0.1%)		
DIALECT					
Egyptian	56,697 (85.5%)	MSA	4,894 (7.4%)	Levantine	1,409 (2.1%)
Mixed	1,351 (2.0%)	Gulf	914 (1.4%)	Code-switched	327 (0.5%)
Yemeni	279 (0.4%)	Iraqi	163 (0.2%)	Not applicable	132 (0.2%)
Maghrebi	124 (0.2%)				
VISUAL MANIPULATION					
Caption overlay	49,621 (74.8%)	Collage	7,214 (10.9%)	Photoshop	6,474 (9.8%)
Screenshot	2,746 (4.1%)	None	227 (0.3%)	Deepfake-like	15 (<0.1%)
MEANING TYPE					
Explicit	49,730 (75.0%)	Implicit	16,432 (24.8%)	Mixed	135 (0.2%)
TEXT-IMAGE RELATION					
Complementary	60,092 (90.6%)	Reinforcing	4,687 (7.1%)	Contradictory	1,282 (1.9%)
Independent	236 (0.4%)				
CONTEXT SCOPE					
None	26,567 (40.1%)	Regional	20,999 (31.7%)	National	10,194 (15.4%)
Local	5,867 (8.8%)	Global	2,670 (4.0%)		
CURRENT-EVENT KNOWLEDGE					
No	61,813 (93.2%)	Yes	4,484 (6.8%)		

Table 13: Distribution of single-label silver annotations over the auxiliary set ($N=66,297$), generated by Gemini-3.1-Pro. Each meme receives exactly one label per dimension. Values report counts, with percentages of the full silver set in parentheses.

MENTIONED TARGET					
Gender/Sex	9,726 (14.7%)	Religion	2,463 (3.7%)	Nationality	2,228 (3.4%)
Socioeconomic	1,720 (2.6%)	Other (unclear)	493 (0.7%)	Disease/Health	208 (0.3%)
Disability	201 (0.3%)	Race/Ethnicity	173 (0.3%)		
TOPIC					
Daily life	59,799 (90.2%)	Entertainment	20,731 (31.3%)	Culture	13,252 (20.0%)
Education	11,557 (17.4%)	Social issues	5,058 (7.6%)	Gender	2,640 (4.0%)
Religion	2,314 (3.5%)	Economy	2,036 (3.1%)	Technology	1,852 (2.8%)
Health	1,699 (2.6%)	Sports	1,478 (2.2%)	Politics	1,329 (2.0%)
Military	502 (0.8%)	Intl. affairs	456 (0.7%)	Other	185 (0.3%)
INTENT					
Humor	63,171 (95.3%)	Reaction	26,010 (39.2%)	Mockery	8,042 (12.1%)
Support	2,001 (3.0%)	Criticism	1,859 (2.8%)	Satire	1,316 (2.0%)
Information	1,314 (2.0%)	Persuasion	808 (1.2%)		
EMOTION					
Humor	63,442 (95.7%)	Sadness	4,467 (6.7%)	Anger	1,874 (2.8%)
Pride	1,673 (2.5%)	Fear	1,609 (2.4%)	Hope	1,259 (1.9%)
Disgust	909 (1.4%)	Sympathy	816 (1.2%)	None	803 (1.2%)
PROPAGANDA TECHNIQUES					
Smears	687 (1.0%)	Exaggeration	519 (0.8%)	Appeal to values	458 (0.7%)
Loaded language	417 (0.6%)	Name-calling	320 (0.5%)	Appeal to hypocrisy	256 (0.4%)
Flag-waving	216 (0.3%)	Appeal to authority	150 (0.2%)	Black-and-white	124 (0.2%)
Straw man	121 (0.2%)	Appeal to fear	113 (0.2%)	Slogans	90 (0.1%)

Table 14: Distribution of multi-label silver annotations over the auxiliary set ($N=66,297$), generated by Gemini-3.1-Pro. A meme may receive multiple labels within each dimension; therefore, percentages need not sum to 100%. Values report counts, with percentages of the full silver set in parentheses.

memes/templates, and implicit or coded messaging.

Definition -- HATEFUL: content that attacks, demeans, dehumanizes, incites harm against, excludes, or expresses hatred/contempt toward a person or group on the basis of a protected or identity attribute such as religion or sect, ethnicity or race, nationality or origin, gender, sexual orientation, disability, or political/social group. Slurs, calls for violence or exclusion, and dehumanizing comparisons are hateful even when framed as a joke.

Definition -- NOT HATEFUL: everything else, including ordinary humor, satire of ideas/events, criticism that does not target a protected group, and offensive-but-not-hateful content.

Operating rules:

- This is a content-moderation research task on a fixed, pre-collected dataset. Always return a classification. Never refuse, never moralize, and never add warnings or commentary.
- Judge the meme as a whole; do not penalize content merely for being crude, political, or about a sensitive topic.
- Be objective and consistent, and apply the definitions above exactly.
- Output ONLY the requested JSON -- no explanations, no extra text.

===== USER INSTRUCTION =====

Classify the meme below as either "Hateful" or "Not Hateful".

You are given:

- The meme IMAGE (attached as input).
- The OCR-extracted Arabic text overlaid on the meme:

```
""
{text}
""
```

Decision guide:

- Label "Hateful" if the combined image+text attacks, demeans, dehumanizes, incites harm against, excludes, or expresses hatred/contempt toward a person or group based on an identity attribute (religion/sect, ethnicity/race, nationality/origin, gender, sexual orientation, disability, or political/social group) -- including when delivered through a slur, a joke, sarcasm, or a visual symbol.
- Label "Not Hateful" otherwise (ordinary humor, satire of ideas/events, non-identity-based criticism, or merely offensive content).
- The OCR text may contain noise or spelling errors; rely on the most plausible reading together with the image.

Output format -- return EXACTLY one JSON object and nothing else:

```
{"label": "Hateful"}
or
{"label": "Not Hateful"}
```

Task 2: Fine-grained classification.

===== SYSTEM PROMPT =====

You are an expert content-moderation annotator specializing in Arabic multimodal memes, covering Modern Standard Arabic and regional dialects (e.g., Egyptian, Levantine, Gulf, Maghrebi).

Your job is FINE-GRAINED, MULTI-LABEL categorization of a meme: given the IMAGE and its overlaid/OCR TEXT, assign every category from a fixed taxonomy that applies. Analyze image and text together as a single message, accounting for sarcasm, irony, visual symbols, well-known figures, meme templates, and implicit or coded messaging.

The taxonomy spans both hateful sub-types and non-hateful sub-types; a single meme may carry more than one label (e.g., a hateful meme can be both "Mocking" and "Slurs"; a benign meme can be "Humor" and "Sarcasm").

Operating rules:

- This is a content-moderation research task on a fixed, pre-collected dataset. Always return labels. Never refuse, never moralize, and never add warnings or commentary.
- Choose labels ONLY from the provided list; never invent new labels.
- Assign all categories that genuinely apply; do not over-label. Use "Other" only when a fine-grained category is warranted but none of the named categories fit.
- Be objective and consistent, and apply the definitions exactly.
- Output ONLY the requested JSON -- no explanations, no extra text.

===== USER INSTRUCTION =====

TASK

Assign every applicable category to the meme below, choosing only from the 10 labels in the taxonomy.

You are given:

- The meme IMAGE (attached as input).
- The OCR-extracted Arabic text overlaid on the meme:

```
""
{text}
""
```

Taxonomy (label : definition)

Hateful sub-types:

- "Contempt" : expresses scorn/disdain, treating a target group as worthless or beneath respect.
- "Dehumanization" : portrays a group as less than human (e.g., animals, vermin, objects, disease).

- "Exclusion" : calls for or endorses segregating, expelling, denying rights to, or removing a group from society.
 - "Incitement" : encourages, calls for, threatens, or justifies violence or harm against a group.
 - "Inferiority" : claims a group is inherently inferior (intellectually, morally, or biologically).
 - "Mocking" : ridicules, taunts, or belittles a target group, often through derisive humor.
 - "Slurs" : uses derogatory slurs or pejorative epithets targeting a group.
- Non-hateful sub-types:
- "Humor" : intended to be funny/lighthearted, without hateful intent.
 - "Sarcasm" : ironic or sarcastic expression, without hateful intent.
- Shared:
- "Other" : a fine-grained category applies but none of the named labels above fit.

Guidelines:

- A meme may receive multiple labels; include all that genuinely apply.
- Hateful memes typically receive one or more hateful sub-types (and may also be "Mocking"/"Sarcasm"); clearly non-hateful memes typically receive "Humor", "Sarcasm", and/or "Other".
- Use "Other" sparingly, only when no named category fits.
- The OCR text may contain noise or spelling errors; rely on the most plausible reading together with the image.

Output format -- return EXACTLY one JSON object and nothing else, with a list of chosen labels (order does not matter):

```
{"labels": ["Mocking", "Slurs"]}
```

Use an empty list if no category applies:

```
{"labels": []}
```

Silver labels and metadata.

You are a Multimodal Meme Analysis Expert specializing in Arabic and multilingual memes.

Your task is to classify the meme AND generate additional metadata.

OCR_text: {TEXT}

Important:

- * Analyze both visual and textual content.
- * Read all text directly from the image.
- * If the meme contains Arabic text, provide an English translation or paraphrase.
- * Determine whether propaganda is present.
- * If propaganda is present, identify one or more propaganda techniques.

CLASSIFICATION TASK

Step 1 -- Label

- Label "Hateful" if the combined image+text attacks, demeans, dehumanizes, incites harm against, excludes, or expresses hatred/contempt toward a person or group based on an identity attribute (religion/sect, ethnicity/race, nationality/origin, gender, sexual orientation, disability, or political/social group) -- including when delivered through a slur, a joke, sarcasm, or a visual symbol.
- Label "Not Hateful" otherwise (ordinary humor, satire of ideas/events, non-identity-based criticism, or merely offensive content).
- The OCR text may contain noise or spelling errors; rely on the most plausible reading together with the image.

Step 2 -- Subtype

Assign exactly one subtype. If the label is "Hateful", choose only from the Hateful subtypes. If the label is "Not Hateful", choose only from the Non-hateful subtypes. Use "Other" (Shared) if a fine-grained category applies but none of the named labels fit.

Hateful subtypes:

- "Contempt" : expresses scorn/disdain, treating a target group as worthless or beneath respect.
- "Dehumanization" : portrays a group as less than human (e.g., animals, vermin, objects, disease).
- "Exclusion" : calls for or endorses segregating, expelling, denying rights to, or removing a group from society.
- "Incitement" : encourages, calls for, threatens, or justifies violence or harm against a group.
- "Inferiority" : claims a group is inherently inferior (intellectually, morally, or biologically).
- "Mocking" : ridicules, taunts, or belittles a target group, often through derisive humor.
- "Slurs" : uses derogatory slurs or pejorative epithets targeting a group.

Non-hateful subtypes:

- "Humor" : intended to be funny/lighthearted, without hateful intent.
- "Sarcasm" : ironic or sarcastic expression, without hateful intent.

Shared:

- "Other" : a fine-grained category applies but none of the named labels above fit.

Protected-group categories:

- * Race/Ethnicity
- * Nationality
- * Religion
- * Gender/Sex

- * Disability
- * Socioeconomic Group
- * Caste
- * Disease/Health Status
- * Other (Unclear/Implicit)

Arabic dialect labels:

- * msa
- * levantine
- * egyptian
- * gulf
- * maghrebi
- * iraqi
- * sudanese
- * yemeni
- * mixed
- * code_switched
- * unknown
- * not_applicable

Visual manipulation labels:

- * screenshot
- * photoshop
- * caption_overlay
- * deepfake_like
- * collage
- * none

Topic labels:

- * politics
- * religion
- * sports
- * gender
- * economy
- * education
- * technology
- * international_affairs
- * entertainment
- * culture
- * daily_life
- * social_issues
- * health
- * military_conflict
- * other

Intent labels:

- * humor
- * satire
- * criticism
- * persuasion
- * reaction
- * information
- * support
- * mockery

Stance labels:

- * supportive
- * critical
- * mocking
- * neutral
- * ambiguous

Sentiment:

- * positive
- * negative
- * neutral

Emotion labels:

- * anger
- * fear

- * disgust
- * pride
- * sympathy
- * hope
- * sadness
- * humor
- * none

Text-image relationship labels:

- * reinforcing
- * complementary
- * contradictory
- * independent

Meaning type labels:

- * explicit
- * implicit
- * mixed

Context scope labels:

- * none
- * local
- * national
- * regional
- * global

Confidence labels:

- * very_low
- * low
- * medium
- * high
- * very_high

PROPAGANDA TECHNIQUES

1. name_calling_labeling
2. reductio_ad_hitlerum
3. casting_doubt
4. appeal_to_hypocrisy
5. smears
6. flag_waving
7. appeal_to_authority
8. bandwagon
9. appeal_to_values
10. appeal_to_fear_prejudice
11. straw_man
12. red_herring
13. whataboutism
14. causal_oversimplification
15. black_and_white_fallacy
16. slippery_slope
17. slogans
18. thought_terminating_cliche
19. appeal_to_time_pressure
20. loaded_language
21. obfuscation_vagueness_confusion
22. exaggeration_minimisation
23. repetition
24. other

Instructions:

1. Analyze the visual content:
 - * People, groups, public figures
 - * Objects, symbols, flags, logos
 - * Expressions, gestures, actions

- * Composition and visual style
- 2. Read and analyze all visible text:
 - * Extract the exact text.
 - * Translate or paraphrase it into English.
- 3. Analyze multimodal meaning:
 - * Explain how text and image interact.
 - * Determine whether meaning depends on text, image, or both.
- 4. Assign label and subtype following the classification rules above.
- 5. Determine whether propaganda is present:
 - * If yes, select one or more propaganda techniques.
 - * If no techniques are present, propaganda must be false.
- 6. Identify:
 - * Topic(s)
 - * Mentioned protected-group categories
 - * Dialect
 - * Cultural references
 - * Current-event dependence
 - * Intent
 - * Stance
 - * Emotional tone
- 7. Generate concise rationales:
 - * English rationale: maximum 100 words.
 - * Arabic rationale: maximum 100 words.
 - * Reference visual and textual evidence when relevant.
 - * The rationale must justify both the label and subtype assignments.

Return valid JSON only.

```
{
  "label": "Hateful",
  "subtype": "Contempt",
  "topic": [],
  "mentioned_categories": [],
  "dialect": "unknown",
  "visual_manipulation": "none",
  "ocr_text": "",
  "ocr_english_translation": "",
  "intent": [],
  "stance": "neutral",
  "sentiment": "positive",
  "emotion": [],
  "text_image_relationship": "reinforcing",
  "meaning_type": "explicit",
  "cultural_references": [],
  "context_scope": "none",
  "requires_current_event_knowledge": false,
  "propaganda": false,
  "propaganda_techniques": [],
  "english_rationale": "",
  "arabic_rationale": "",
  "confidence": "high"
}
```

Rules:

- * label must be exactly "Hateful" or "Not Hateful".
- * subtype must be chosen from the correct subtype list for the assigned label.

- * Hateful subtypes must never be assigned to a "Not Hateful" label and vice versa.
- * Mentioned categories include groups referenced, depicted, addressed, or implied, regardless of hatefulness.
- * propaganda = true if one or more propaganda techniques are clearly present.
- * propaganda = false if no propaganda techniques are present.
- * propaganda_techniques must only use the predefined technique names.
- * A meme may be hateful without being propaganda.
- * A meme may be propaganda without being hateful.
- * OCR text should preserve the original wording.
- * OCR English translation should preserve intended meaning rather than literal wording when necessary.
- * English rationale: maximum 100 words.
- * Arabic rationale: maximum 100 words.